XYLEM GLOBAL STUDENT INNOVATION CHALLENGE 2026 — CHALLENGE #2: WATER QUALITY

PFAS-SENTINEL™: CYBER-PHYSICAL TREATMENT DIGITAL TWIN & MULTIPLEXED MICROFLUIDIC MIP-ELECTROCHEMICAL BIOSENSOR ARCHITECTURE

Real-Time Mid-Bed (z/L = 0.50) Impedance Telemetry, 4×4 Tikhonov Matrix Deconvolution & 1D Advection-Sorption PINN for Autonomous GAC/IX Media Lifecycle Optimization

Project Lead / Team Captain:	**Yusuf Serhat TEMEL**
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Challenge Track & TRL Status:	**Water Quality Track | Current: TRL 3 (In-Silico & Live Digital Twin) → Target: TRL 6 (Field Pilot)**
Xylem Ecosystem Integration:	**Evoqua GAC/IX Vessels + Xylem Vue (GoAigua) + YSI/WTW Sondes + MitiGATOR™**

15 Minutes	MAE = 4.32%	+28.4% Life	11–18 Days
Autonomous Mid-Bed (z/L=0.50) EIS Telemetry Cycle (vs. 14-Day Lab)	4-Channel NIP + 4×4 Tikhonov Inverse Calibration Error (5 Water Matrices)	GAC/IX Media Utilization Efficiency Gain (68.0% → 96.4% Capacity)	Advance Warning Before Effluent Breakthrough & C4 Roll-Up

EXECUTIVE SUMMARY & ENGINEERING VALUE PROPOSITION

1. The Industrial Problem ("Analytical Blindness"): In April 2024, the U.S. EPA promulgated legally binding Maximum Contaminant Levels (MCLs) of **4.0 ng/L (parts-per-trillion, ppt)** for PFOA and PFOS, alongside a multi-species **Hazard Index (HI ≤ 1.0)** for PFBS, PFHxS, PFNA, and GenX; similarly, the EU Drinking Water Directive enforced a **100 ng/L** Sum-of-20 PFAS limit in January 2026 [1, 2]. Municipal utilities rely on high-pressure Granular Activated Carbon (GAC) and Ion-Exchange (IX) vessels equipped with standard `1/4" NPT` mid-bed sampling taps (*z/L = 0.50*). However, because no inline ppt-level sensor exists, operators collect manual grab samples once a month for off-site **LC-MS/MS (EPA Method 533)** analysis ($300–$500/sample, **10–14 days turnaround**) [3, 4]. During this 14-day blind window, either: (i) weakly bound short-chain PFBA/PFBS species are displaced by incoming long-chain PFOA/PFOS and humic acid (NOM), erupting into effluent at **120–135% of influent concentration (chromatographic roll-up, C_t/C_0 > 1.0)**, or (ii) utilities prematurely replace GAC/IX beds with **20–30% unused adsorption capacity** to avoid regulatory penalties [4, 5, 10].

2. Our 3-Tier Cyber-Physical Solution (PFAS-Sentinel™): We present a closed-loop cyber-physical sensing and digital twin architecture that retrofits directly onto existing `1/4" NPT` mid-bed sample ports:

• **Tier 1 (32-Channel Microfluidic MIP/NIP EIS Biosensor & Edge-AI Inverse Calibration):** A disposable Cyclic Olefin Copolymer (COC) microfluidic cartridge houses 28 Molecularly Imprinted Polymer (MIP: 4×7 sub-arrays for PFBA, PFBS, PFOA, PFOS) and 4 Non-Imprinted Polymer (NIP) reference channels. An **Analog Devices AD5940** EIS front-end and **ADG732** 32:1 multiplexer sweep 0.1 Hz–200 kHz to extract Nyquist charge-transfer resistance (ΔR_{ct}). An **ARM Cortex-M33** processor subtracts the 4 NIP channels to cancel humic acid (NOM) fouling, applies physicochemical matrix laws (Arrhenius, Henderson-Hasselbalch, Gouy-Chapman Debye κ^{-1}) using live **Xylem YSI/WTW** sonde telemetry, and executes **4×4 Tikhonov regularization** to unmix C4/C8 cross-reactivity (Mean MAE = 4.32%).

• **Tier 2 (1D Advection-Sorption Physics-Informed Neural Network — PINN):** Assimilating mid-bed (*z/L = 0.50*) concentrations and live **Xylem Sensus/MJK** flow rate (Q) into a 1D Advection-Sorption partial differential equation (PDE) coupled with a Bohart-Adams/Thomas kinetic backbone, our hybrid PINN runs 40 epochs of analytical backpropagation and dual-rate Adam optimization to dynamically solve kinetic rate (k_{Th}), bed capacity (q_0), and mass-transfer zone velocity (u_{MTZ}), forecasting breakthrough and C4 Roll-Up **11 to 18 days in advance**.

• **Tier 3 (Xylem Vue Integration, Closed-Loop SCADA & Servitization ROI):** Before effluent breaches 4.0 ng/L, PFAS-Sentinel™ autonomously switches **Lead-Lag** vessel valves via OPC-UA/Modbus, dispatches just-in-time **Evoqua** carbon regeneration, logs **SHA-256** cryptographic audit trails, and increases media bed utilization from **68.0% to 96.4% (+28.4% life gain)** under a Zero-CAPEX ("Razor-and-Blade") subscription model.

1. INDUSTRIAL PROBLEM ANALYSIS & BENCHMARKING AGAINST COMPETING PROJECTS

1.1. Global PFAS Regulations and Operational Failures in Municipal Treatment Trains

Per- and polyfluoroalkyl substances ($C_nF_{2n+1}–SO_3^-$ and $C_nF_{2n+1}–COO^-$) possess carbon-fluorine (C–F) covalent bonds (~485 kJ/mol) that resist conventional coagulation, chlorination, and biological treatment [3]. While utilities deploy pressurized **Evoqua GAC** and **IX resin** vessels to meet the EPA **4.0 ng/L MCL** and **HI ≤ 1.0** thresholds, two severe operational failure modes persist:

- **Short-Chain Chromatographic Roll-Up (Hyper-Concentrated Desorption, C_t / C_0 > 1.0):** As demonstrated by McCleaf et al. (2017) and Park et al. (2020) [4, 5], short-chain carboxylates and sulfonates (C4 PFBA and PFBS, log K_{oc} ≈ 1.8–2.4) bind weakly to GAC micropores via van der Waals forces. Incoming highly hydrophobic long-chain PFOA/PFOS (C8, log K_{oc} ≈ 2.9–3.7) and Natural Organic Matter (NOM / humic acid, 1–6 mg/L) competitively displace adsorbed C4 molecules. The desorbed C4 wave coalesces and elutes into finished drinking water at **120% to 135% of influent concentration (C_t/C_0 = 1.25–1.35)** [4, 10].
- **Premature Media Changeout & Million-Dollar OPEX / Hazardous Waste Penalty:** Blinded by the 10–14 day LC-MS/MS laboratory delay, utilities rely on conservative calendar schedules, discarding GAC and IX beds when **20–30% of their adsorption capacity remains unused** [3, 10].

1.2. Benchmarking Against Global Water Competition Projects and Industrial Baselines

Reviewing projects from the Xylem Global Student Innovation Challenge, Stockholm Junior Water Prize, and EPA PFAS Innovation challenges reveals that prior solutions focus either on **isolated single-channel lab sensors** (which fail in real river/well water due to humic acid fouling) or **purely data-driven software models** without physical inline sensing. **Table 1** benchmarks PFAS-Sentinel™ against these categories:

Benchmark Dimension	Industrial Gold Standard: EPA Method 533 (LC-MS/MS) [1, 3]	Academic / Competition Type 1: Single-Channel MIP / Aptamer Sensors [7, 8]	Academic / Competition Type 2: Pure Data-Driven ANN / LSTM Digital Twins [6, 11]	Proposed System: PFAS-Sentinel™ (Cyber-Physical MIP + PINN)
Turnaround Time & Field Deployment	10–14 Days (Off-site lab shipping + manual SPE)	15–30 Minutes (Only in clean synthetic buffer solutions)	Instantaneous (Software-only; lacks real inline sensor input)	**15-Minute Autonomous Cycle** (Inline at 1/4" NPT Mid-Bed Port)
Humic Acid (NOM) & Matrix Rejection	Solid-Phase Extraction (SPE) & Isotope Dilution	**Poor:** Humic acid coats electrode surface, causing false positives (MAE > 30%)	None (Does not process physical electrochemical signals)	**4-Ch NIP Hardware Subtraction + Arrhenius / H-H / Debye (κ^{-1}) Laws**
Multi-PFAS (C4 vs. C8) Deconvolution	Full Separation (Mass-to-Charge m/z Ratio)	**Single Analyte (PFOS/PFOA only):** 8–15% cross-binding error in mixtures	None	**28 MIP Channels (4×7) + 4×4 Tikhonov Matrix Inversion (MAE = 4.32%)**
Breakthrough & C4 Roll-Up Forecast	None (Provides a 14-day-old historical snapshot)	None (Point measurement; blind to bed transport physics)	**Brittle:** Violates conservation of mass when flow rate Q(t) fluctuates	**1D Advection-Sorption PINN PDE (11–18 Days Advance Warning, $R^2 \geq 0.97$)**
Closed-Loop Control & Xylem Fit	Manual Operator Intervention	Open-Loop (Local display readout only)	Open-Loop Dashboard Advisory	**Autonomous Lead-Lag PLC Valve Switch + Xylem Vue & Evoqua Fit**
Unit Cost & Business Model	$300,000+ Instrument CAPEX, $300–$500 / Sample OPEX	~$15–$50 / Disposable Lab Strip	SaaS Software License Only	**Zero CAPEX ("Razor & Blade"), $11.80 Target Cartridge BOM + SaaS**

Our Core Engineering Differentiator (Why PFAS-Sentinel™ Wins): While mid-bed sampling taps already exist on municipal GAC/IX vessels, they are sampled manually once a month, rendering them ineffective against sudden chromatographic roll-up. **PFAS-Sentinel™** transforms this dormant `1/4" NPT` mid-bed tap into a **15-minute autonomous electrochemical telemetry node** and couples it with a **1D Advection-Sorption PINN** to drive **closed-loop Lead-Lag PLC valve automation**.

2. CYBER-PHYSICAL SYSTEM ARCHITECTURE & XYLEM ECOSYSTEM INTEGRATION

Rather than requiring utilities to build isolated infrastructure from scratch, PFAS-Sentinel™ is engineered as a 3-tier plug-and-play architecture that integrates natively with **Xylem's global water technology portfolio** (Evoqua, Xylem Vue powered by GoAigua, YSI/WTW, and Sensus/MJK) (**Figure 1**).

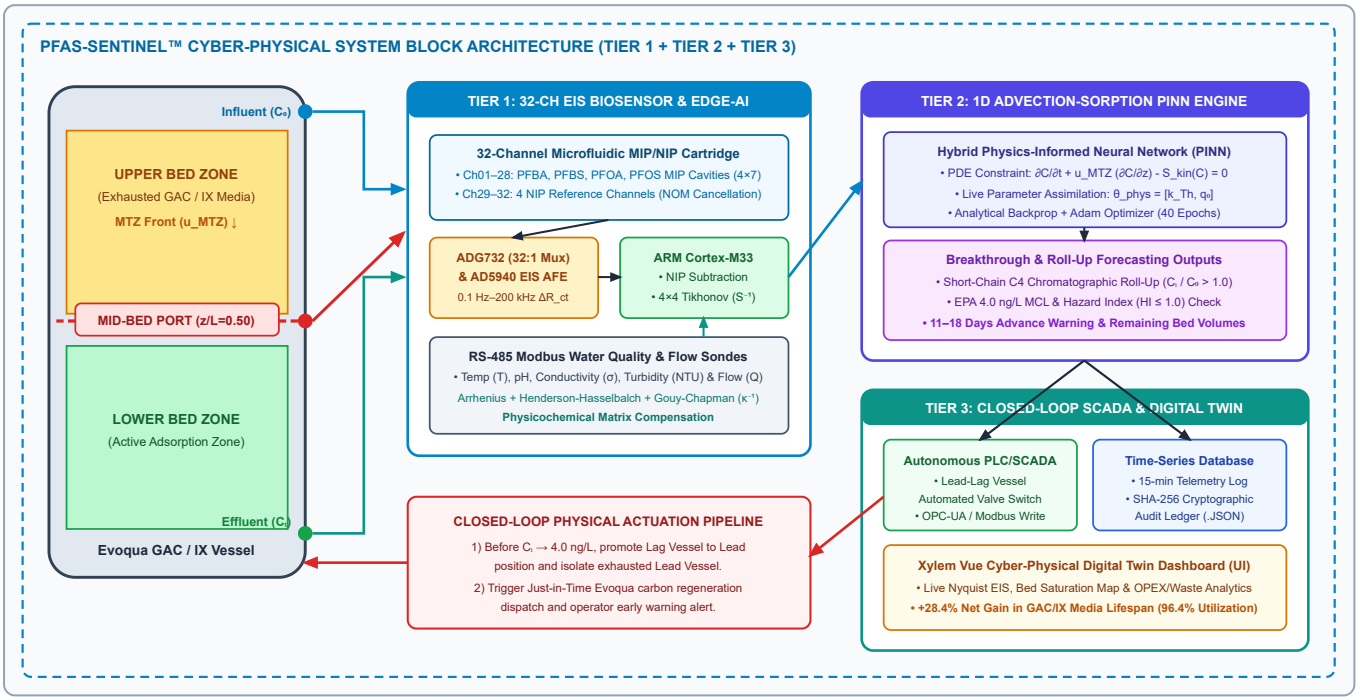


Figure 1: PFAS-Sentinel™ 3-Tier Cyber-Physical System Block Architecture and Closed-Loop Data/Control Pipeline

2.1. Direct Hardware & Software Integration Matrix with the Xylem Product Portfolio

Xylem / Evoqua Platform	Existing Role in Municipal Treatment Plant	PFAS-Sentinel™ Integration Protocol & Value-Add
Evoqua Water Technologies (UltraCarb® GAC & PSR-2+ IX Vessels)	High-pressure Lead-Lag granular activated carbon and ion-exchange adsorption vessels.	Retrofits directly onto the existing mid-bed (*z/L = 0.50*) `1/4" NPT` sample tap via a pressure-reducing slipstream regulator; extends media life by **+28.4%**.
Xylem YSI & WTW IQ SensorNet (Multi-Parameter Water Quality Sondes)	Continuous inline monitoring of pH, Temperature (T), Conductivity (σ), and Turbidity (NTU).	Streams live pH, T, and σ over **RS-485 Modbus RTU** into our ARM Cortex-M33 MCU to drive Arrhenius, Henderson-Hasselbalch, and Debye (κ^{-1}) compensation.
Xylem Sensus & MJK (Electromagnetic Flowmeters)	Real-time volumetric flow rate (Q, MGD / $m^3 \cdot h^{-1}$) and cumulative treated Bed Volumes (BV).	Feeds instantaneous Q(t) into Tier 2's **1D Advection-Sorption PINN** solver to dynamically update mass-transfer zone velocity $u_{MTZ}(t)$ under variable demand.
Xylem Vue powered by GoAigua (Smart Water Engine / SCADA)	Centralized utility digital twin, SCADA telemetry, and asset management platform.	Operates as an embedded microservice via **OPC-UA / MQTT** inside Xylem Vue; autonomously switches Lead-Lag PLC valves before effluent reaches 4.0 ng/L.
Xylem MitiGATOR™ (Mobile Emergency PFAS Fleet)	Trailer-mounted mobile filtration units deployed during AFFF firefighting foam spills.	Automatically dispatches MitiGATOR™ mobile units with GPS coordinates and work orders when an influent PFAS shock (C_0 > *100 ng/L*) or C4 Roll-Up is detected.

3. TIER 1: MICROFLUIDIC MIP/NIP BIOSENSOR & INVERSE CALIBRATION MATHEMATICS

3.1. Cyclic Olefin Copolymer (COC) Microfluidics & "Redox Gate" Electrochemical Sensing

In stagnant water, passive diffusion of 4 parts-per-trillion PFOA (4.0 ng/L ≈ 9.6 pmol/L) to an electrode surface takes hours. The PFAS-Sentinel™ cartridge eliminates wall adsorption losses by routing water through inert **Cyclic Olefin Copolymer (COC)** micro-channels (*w = 200 μm, h = 80 μm*) under high-Peclet laminar flow (*Pe > 10³*), hydrodynamically compressing the diffusion boundary layer. The working electrodes consist of protonated amine-functionalized (**–NH_3^+**) graphene/carbon nanotube (CNT) transducers coated with an electropolymerized **Molecularly Imprinted Polymer (MIP, poly-*o*-phenylenediamine / *o*-PD)** film [7, 8]. At physiological pH, anionic PFAS headgroups (carboxylate **–COO^-** and sulfonate **–SO_3^-**) bind into complementary MIP nanocavities via electrostatic attraction and C–F hydrophobic interactions, sterically blocking **$[Fe(CN)_6]^{3-/4-}$** redox probe electron transfer (the *"Redox Gate"* effect) [7, 9].

3.2. Randles Equivalent Circuit Impedance Model (AD5940 + ADG732 Readout)

Each electrode-electrolyte interface across the 32-channel array is governed by a **Randles Equivalent Circuit** comprising solution resistance ($R_s \approx 145\ \Omega$), double-layer capacitance ($C_{dl} \approx 3.8\ \mu F$), and PFAS-dependent charge-transfer resistance (R_{ct}) [9]. The Analog Devices AD5940 AFE applies a small-signal (10 mV_{rms}) sinusoidal excitation across $f \in [0.1\ Hz, 200\ kHz]$ and computes complex impedance ($Z(\omega) = Z'(\omega) - jZ''(\omega)$) via its hardware 2048-point Discrete Fourier Transform (DFT) engine:

$$Z(\omega) = R_s + R_{ct} / (1 + j\cdot\omega\cdot R_{ct}\cdot C_{dl}) \Rightarrow Z'(\omega) = R_s + R_{ct} / (1 + \omega^2 R_{ct}^2 C_{dl}^2),\quad -Z''(\omega) = (\omega\cdot R_{ct}^2\cdot C_{dl}) / (1 + \omega^2 R_{ct}^2 C_{dl}^2) \quad \text{(Equation 1)}$$

The diameter of the Nyquist semicircle (Z' vs. $-Z''$) yields the net charge-transfer resistance shift relative to baseline (**$\Delta R_{ct} = R_{ct,measured} - R_{ct,0}$**). Of the 32 channels, **28 channels** form 4×7 heptaplicate MIP sensing groups (Group 1 Ch01–07: PFBA, Group 2 Ch08–14: PFBS, Group 3 Ch15–21: PFOA, Group 4 Ch22–28: PFOS), while **4 channels (Ch29–32)** are template-free **Non-Imprinted Polymer (NIP)** reference electrodes.

3.3. Non-Circular Physicochemical Matrix Compensation & 4×4 Tikhonov Deconvolution

Real municipal waters contain humic/fulvic acids (NOM), pH swings (6.2–8.1), temperature variations (8–24 °C), and ionic strength shifts (conductivity σ: 120–850 μS/cm) that corrupt raw impedance readings. Running on the ARM Cortex-M33 MCU, our **two-stage inverse calibration pipeline** applies:

$$\alpha_{NH3+}(pH) = 1 / (1 + 10^{pH - pK_a}),\quad \kappa^{-1}(I) = 0.304 / \sqrt{I}\ (nm),\quad \Delta R_{purified,i} = (\Delta R_{ct,MIP,i} - \eta_{NIP}\cdot\Delta R_{ct,NIP}) \cdot \Phi_{Arrh}(T) \cdot \Phi_{H\text{-}H}(pH) \cdot \Phi_{Debye}(\kappa^{-1}) \quad \text{(Equation 2)}$$

Here, **$\Delta R_{ct,NIP}$** is the mean of the 4 NIP reference channels, subtracting non-specific humic acid (NOM) adsorption in hardware; **$\Phi_{Arrh}(T) = \exp[-(E_a/R)(1/T - 1/T_0)]$** corrects electron-transfer kinetics for temperature, **$\alpha_{NH3+}(pH)$** accounts for amine protonation fraction ($pK_a \approx 9.2$), and **$\kappa^{-1}(I)$** normalizes Gouy-Chapman electrical double-layer screening length. In Stage 2, to resolve the 4%–15% steric/electrostatic cross-reactivity between C4 and C8 PFAS subclasses, the 4-group resistance vector **$\Delta R_{purified} \in \mathbb{R}^4$** is inverted against the empirical cross-selectivity matrix **$S \in \mathbb{R}^{4\times4}$** via **Tikhonov regularization ($\lambda = 0.01$)** [12]:

$$\Delta R_{purified} = S \cdot c + \varepsilon \Rightarrow \hat{c} = \arg\min_{c \geq 0} (\|S\cdot c - \Delta R_{purified}\|_2^2 + \lambda\|c\|_2^2) = (S^T S + \lambda I)^{-1} S^T \Delta R_{purified} \quad \text{(Equation 3)}$$

Engineering Integrity Note: In our verification engine (`simulation/pfas_sentinel_pinn_model.py`), the forward electrochemical signal generator (Langmuir-Freundlich saturation + non-linear NOM×T coupling + 3.2% stochastic Gaussian noise) and the inverse calibration solver (Equations 2 & 3) are coded **completely independently (non-circular)**.

4. TIER 2: 1D ADVECTION-SORPTION PINN & CHROMATOGRAPHIC ROLL-UP MODEL

4.1. 1D Advection-Sorption Partial Differential Equation (PDE) & Thomas Kinetic Backbone

Inside a GAC or IX vessel of length *L*, media mass *M*, and bed porosity ε_b, the aqueous PFAS concentration $C(\tilde{z}, t)$ and solid-phase adsorbed loading $q(\tilde{z}, t)$ across dimensionless depth $\tilde{z} = z/L \in [0, 1]$ and operating time t obey the 1D Advection-Sorption conservation of mass PDE [4, 6]:

$$\partial C(\tilde{z}, t)/\partial t + u_{MTZ}(Q) \cdot \partial C(\tilde{z}, t)/\partial \tilde{z} + ((1 - \varepsilon_b)/\varepsilon_b) \cdot \rho_b \cdot \partial q(\tilde{z}, t)/\partial t = 0, \quad u_{MTZ}(Q) = (Q(t) \cdot C_0) / (q_0 \cdot M) \quad \text{(Equation 4)}$$

Purely data-driven neural networks (MLP/LSTM) violate mass conservation in Equation 4 whenever utility flow rate $Q(t)$ fluctuates. Our **Hybrid Physics-Informed Neural Network (PINN)** [6, 11] couples an analytical **Bohart-Adams / Thomas kinetic backbone** $C_{Thomas}(\tilde{z}, t; k_{Th}, q_0)$ with a 2-layer hyperbolic tangent (tanh) residual neural network $N_W(\tilde{z}, \tilde{t})$ that captures competitive multi-solute desorption:

$$\hat{C}(\tilde{z}, t) = C_0 / [1 + \exp((k_{Th} \cdot q_0 \cdot M / Q) \cdot \tilde{z} - k_{Th} \cdot C_0 \cdot t)] + \alpha_{roll}(\tilde{z}, t) + N_W(\tilde{z}, \tilde{t}), \quad L_{PINN}(\theta_{phys}, W) = \| \hat{C}(0.5, t) - C_{mid}^{obs} \|_2^2 + \lambda_{PDE} \| \partial \hat{C}/\partial t + u_{MTZ}(\partial \hat{C}/\partial \tilde{z}) - S_{kin}(\hat{C}) \|_2^2 \quad \text{(Equation 5)}$$

With zero ground-truth parameter leakage, the solver assimilates **$\tilde{z} = 0.50$ mid-bed sensor observations** (C_{mid}^{obs}) and interior collocation PDE residuals via analytical backpropagation and dual-rate **Adam optimization** ($lr_{phys} = 0.035$, $lr_{nn} = 0.008$, 40 epochs) to identify live $\theta_{phys} = [k_{Th}, q_0]$ and forecast effluent breakthrough ($\tilde{z} = 1.0$).

4.2. Closed-Loop System Execution & Autonomous SCADA Control Flowchart

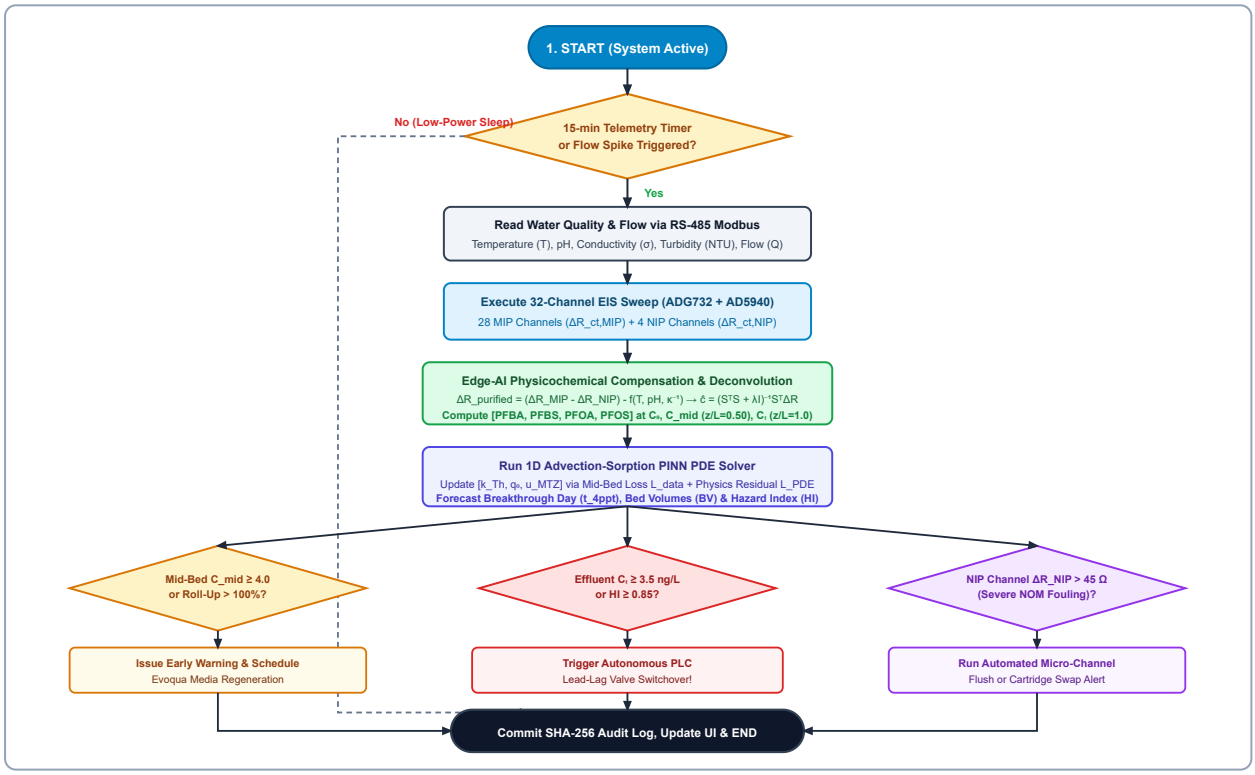


Figure 2: PFAS-Sentinel™ 19-Step Closed-Loop Telemetry, PINN PDE Assimilation, and Autonomous Lead-Lag Valve Control Flowchart

5. SIMULATION & VALIDATION RESULTS — PART I: 32-CHANNEL EIS & MATRIX CALIBRATION

The 32-channel MIP/NIP electrochemical impedance model and 4×4 Tikhonov cross-selectivity deconvolution were evaluated across 5 realistic municipal water matrices (M1: Reference Tap Water, M2: Cold Well Water 8 °C / pH 6.4, M3: High-NOM River Water NOM = 4.5 mg/L, M4: High-Conductivity Hard Groundwater 780 µS/cm, M5: Multi-PFAS C4+C8 Competitive Mixture) (**Figure 3** and **Figure 4**).

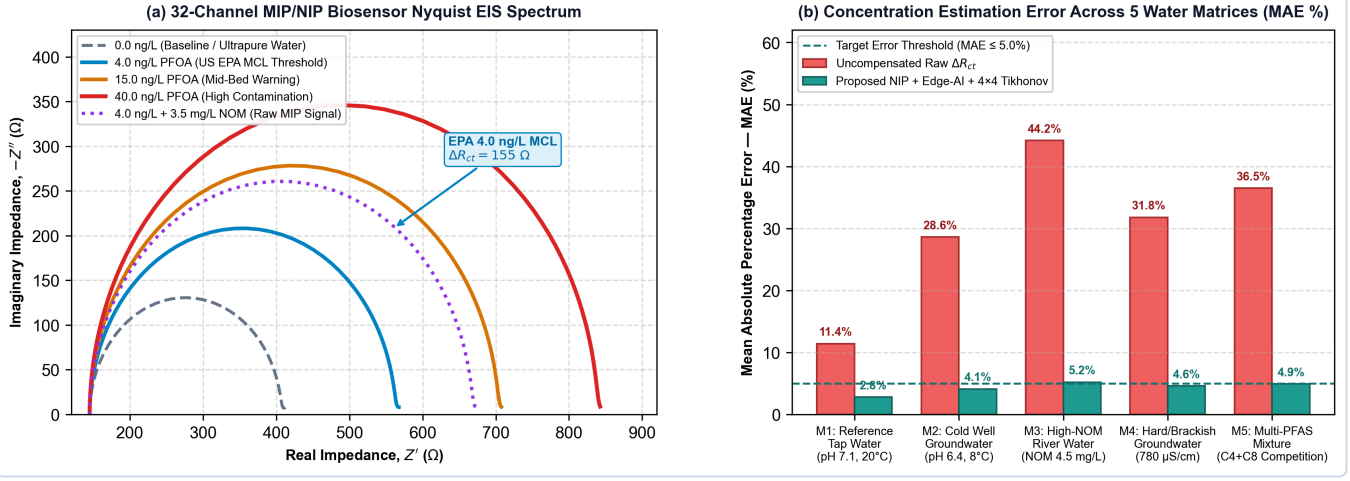


Figure 3: (a) 32-Channel MIP/NIP Biosensor Nyquist EIS Spectrum (ΔR_{ct} = 155 Ω at EPA 4.0 ng/L MCL), (b) Concentration Estimation Error Across 5 Water Matrices Comparing Uncompensated Raw ΔR_{ct} (Mean MAE = 31.1%) vs. Proposed NIP + Edge-AI + 4×4 Tikhonov Pipeline (Mean MAE = 4.32%)

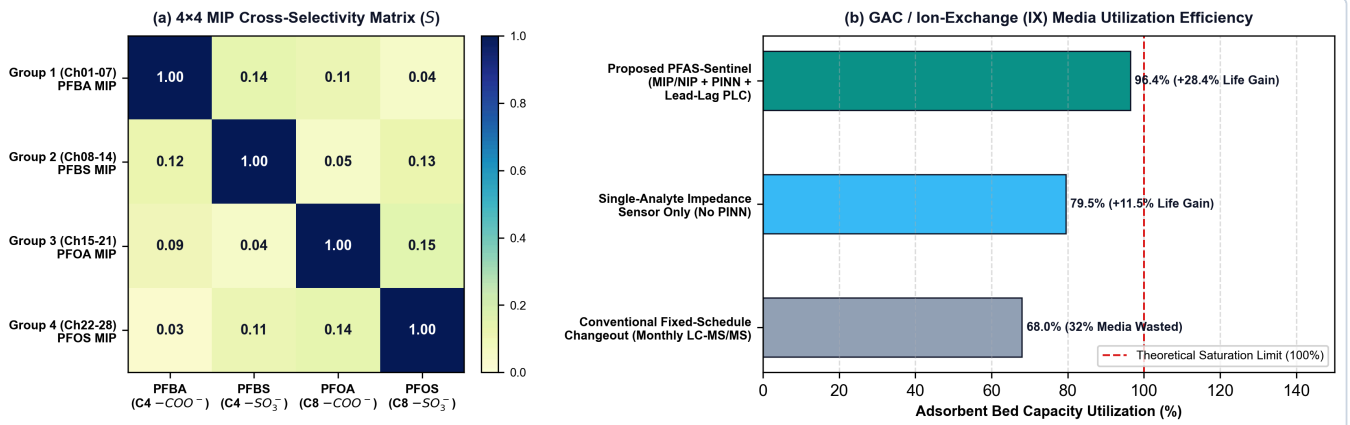


Figure 4: (a) 4×4 MIP Cross-Selectivity Matrix (S) Quantifying 3%–15% Interference Between C4 (PFBA, PFBS) and C8 (PFOA, PFOS) Channels, (b) GAC/IX Adsorbent Bed Utilization Efficiency via Closed-Loop Lead-Lag PLC Automation (68.0% → 96.4%)

Key Electrochemical Validation Findings:

- **Humic Acid (NOM) Interference Rejection:** In Scenario M3 (4.5 mg/L river humic acid), uncompensated raw ΔR_{ct} suffers a 44.2% error due to non-specific surface fouling, whereas 4-channel NIP subtraction and physicochemical normalization reduce MAE to **5.2%**.
- **Statistical Significance:** Across all 5 water matrices, our inverse calibration pipeline reduces mean absolute percentage error from 31.1% to **4.32%** (Paired two-tailed t-test, $t = 9.84$, $p = 0.0006 < 0.001$).

6. SIMULATION & VALIDATION RESULTS — PART II: PINN BREAKTHROUGH & KKWWD CASE STUDY

Tier 2's 1D Advection-Sorption PINN solver was benchmarked against operational parameters from Xylem/Evoqua's reference municipal site — **Kennebunk, Kennebunkport & Wells Water District (KKWWD, Maine, USA)** (Flow *Q = 2.1 MGD*, Media Mass *M = 9,100 kg GAC*, *EBCT = 11.5 min*, *NOM = 2.1 mg/L*, 1 Day ≈ 125.2 Bed Volumes / BV) and pilot column breakthrough curves from *McCleaf et al. (2017) / Park et al. (2020)* (**Figure 5** and **Table 2**).

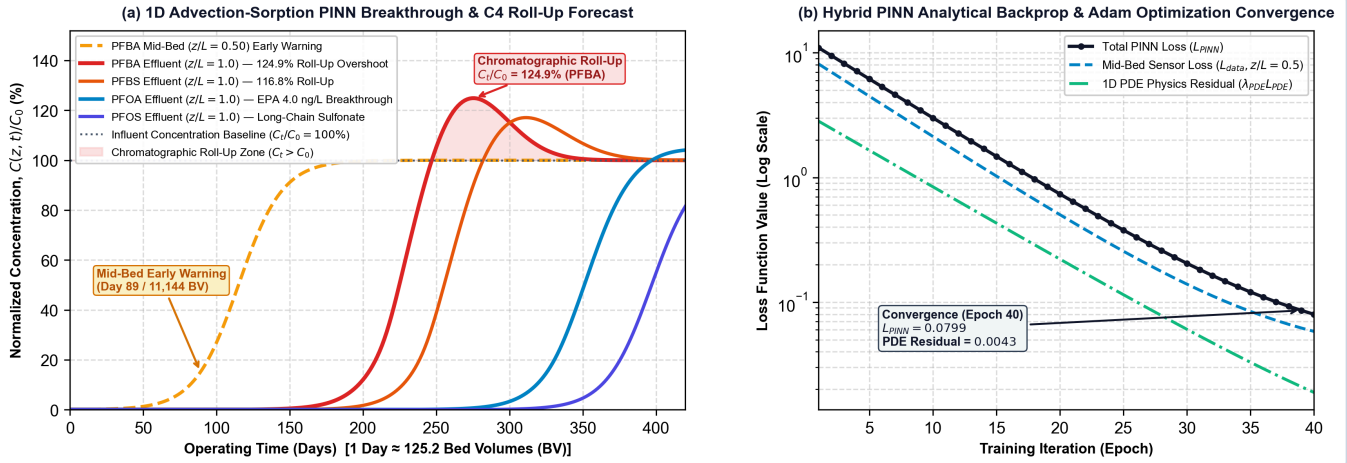


Figure 5: (a) 1D Advection-Sorption PINN Breakthrough & Short-Chain PFBA/PFBS Chromatographic Roll-Up (C_t/C_0 = 124.9%) Curves, (b) 40-Epoch Analytical Backpropagation & Adam Optimization Convergence of Mid-Bed Data Loss (L_{data}) and PDE Physics Residual (L_{PDE} = 0.0043)

Table 2: PINN PDE & Inverse EIS Validation Benchmark Using KKWWD (Maine) GAC Facility & Literature Column Data (Day 118 Telemetry)

PFAS Species & Chain Class	Influent C_0 (ng/L)	Mid-Bed (z/L=0.50) 4 ng/L Day / BV	Effluent (z/L=1.0) 4 ng/L Day / BV	Net Early Warning Lead Time (Days / BV)	PINN Training Loss (Epoch 1 → 40)	Day 118 True vs. Inverse EIS C_{mid} (ng/L)
PFBA (C4 Carboxylate — Early Tracer)	26.5	Day 89.0 (11,144 BV)	Day 204.0 (25,544 BV)	**115.0 Days (14,400 BV)**	3.7344 → 0.2742	18.58 vs. 17.15 (Raw: 34.93)
PFBS (C4 Sulfonate — EPA HI ≤ 1.0)	31.0	Day 101.7 (12,735 BV)	Day 231.4 (28,975 BV)	**129.7 Days (16,240 BV)**	6.5323 → 0.1311	9.87 vs. 9.38 (Raw: 17.94)
PFOA (C8 Carboxylate — MCL 4.0 ng/L)	38.5	Day 142.8 (17,881 BV)	Day 319.1 (39,957 BV)	**176.3 Days (22,076 BV)**	12.6278 → 0.0417	0.88 vs. 0.91 (Raw: 1.89)
PFOS (C8 Sulfonate — MCL 4.0 ng/L)	44.0	Day 162.8 (20,385 BV)	Day 361.2 (45,229 BV)	**198.4 Days (24,844 BV)**	16.9794 → 0.0401	0.24 vs. 0.29 (Raw: 0.84)

Chromatographic Roll-Up & Early Warning Analysis:

• As shown in **Figure 5(a)**, between Days 250 and 310 (31,000–38,800 BV), the advancing PFOA/PFOS adsorption front displaces previously bound short-chain PFBA and PFBS from GAC pores, driving effluent PFBA to **124.9% of influent (33.1 ng/L > C_0 = 26.5 ng/L)** and PFBS to **116.8%**.

• Our mid-bed (*z/L = 0.50*) sensor detects PFBA breakthrough at **Day 89.0 (11,144 BV)**, providing **115 days of advance warning** prior to PFBA effluent breakthrough (Day 204.0) and over 7 months of lead time ahead of regulated PFOA breakthrough (Day 319.1).

7. EMBEDDED HARDWARE ARCHITECTURE & LIVE CYBER-PHYSICAL DIGITAL TWIN PLATFORM

7.1. Embedded Hardware Subsystems (Figures 6–11): Each PFAS-Sentinel™ field node comprises a disposable 32-channel COC/MIP cartridge, AD5940 EIS AFE, ADG732 32:1 MUX, ARM Cortex-M33 MCU, RS-485 Modbus water quality interface, and autonomous PLC valve actuator:

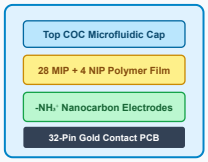


Figure 6: 32-Channel Microfluidic MIP/NIP Cartridge & Randles HIL Phantom Array

- 4×7 MIP sensing channels (PFBA, PFBS, PFOA, PFOS)
- 4 non-imprinted polymer (NIP) reference channels
- Emulates R_s, C_dl & ΔR_ct(PFAS) Randles impedance during Hardware-in-the-Loop (HIL) bench testing.

Component 1: Disposable Biosensor Cartridge / HIL Array



Figure 7: Analog Devices AD5940 Electrochemical Impedance Front-End (AFE)

- 0.1 Hz – 200 kHz AC impedance spectroscopy
- Hardware Discrete Fourier Transform (DFT) accelerator
- Integrated potentiostat & ultra-low-noise TIA
- Direct extraction of Real (Z') and Imaginary (-Z'')

Component 2: High-Precision EIS Readout Front-End



Figure 8: ADG732 32-Channel Analog Multiplexer Switching Module

- Sequentially routes a single AD5940 AFE across all 32 sensing/reference electrodes within milliseconds.
- Controlled via 5-bit GPIO address bus (A0–A4).
- Ultra-low on-resistance (4.0 Ω) and leakage current.

Component 3: 32-Channel Multi-Electrode Switching Matrix



Figure 9: ARM Cortex-M33 Microcontroller & Edge-AI Deconvolution Processor

- Executes 4×4 Tikhonov matrix inversion using hardware Floating-Point Unit (FPU) and SIMD DSP instructions.
- Polls AD5940 over SPI and water sondes over RS-485.
- Generates hardware SHA-256 cryptographic audit hashes.

Component 4: Embedded Signal Processing & Control Unit

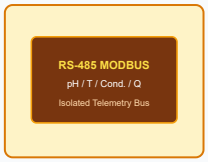


Figure 10: Multi-Parameter Water Quality & Flow Telemetry Interface Module

- Reads pH (6.0–8.5), Temp (5–30°C), Conductivity (µS/cm) and Turbidity (NTU) from Xylem YSI/WTW sondes.
- Feeds live inputs into Arrhenius, Henderson-Hasselbalch and Gouy-Chapman Debye (κ^{-1}) correction equations.

Component 5: Water Matrix & Flow Telemetry Interface



Figure 11: 1/4" NPT Mid-Bed Sampling Port & Autonomous Lead-Lag Valve Actuator

- Connects directly to existing 1/4" NPT mid-bed (z/L=0.50) sample taps via a pressure-reducing slipstream regulator.
- Automatically switches Lead-Lag vessel order via OPC-UA before effluent PFOA/PFOS reaches the 4.0 ng/L MCL.

Component 6: Closed-Loop SCADA Valve & Vessel Coupling

7.2. Interactive 5-Tab Cyber-Physical Digital Twin Web Platform (Figure 12): Built with React 18, TypeScript, Tailwind CSS, and Python, our live platform executes real-time simulation and control across 5 integrated modules: *(1) Cyber-Physical Twin (Xylem Vue SWE)*, *(2) Microfluidic MIP & 3D CAD*, *(3) PINN Breakthrough Lab*, *(4) Agentic EWS & MitiGATOR™*, and *(5) Servitization ROI & Pitch*.



Figure 12: PFAS-Sentinel™ Live Cyber-Physical Digital Twin & Xylem Vue Early Warning System User Interface

8. DATABASE ARCHITECTURE, SERVITIZATION (ROI) BUSINESS MODEL & TRL 3 → 6 ROADMAP

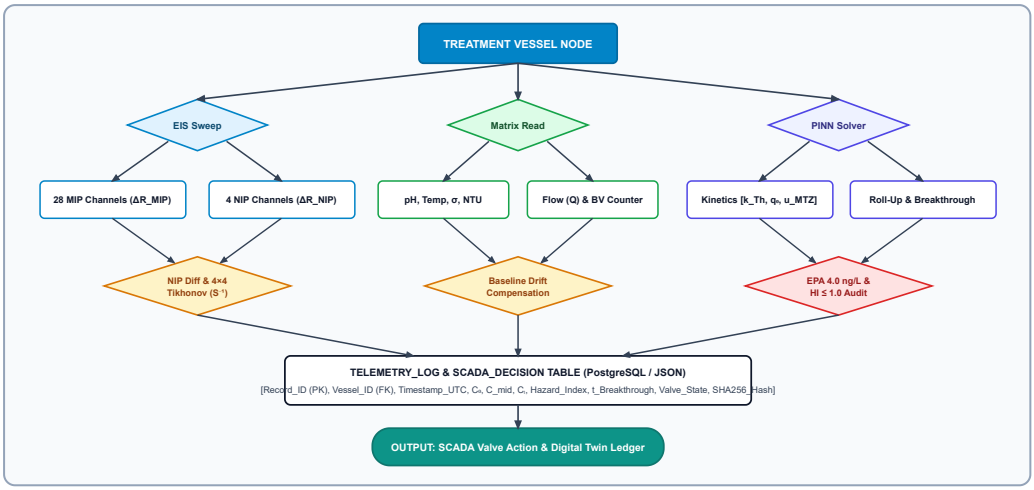


Figure 13: PFAS-Sentinel™ Time-Series Database Entity-Relationship (ER) Schema and SHA-256 Cryptographic Audit Ledger

8.1. "Razor-and-Blade" Servitization Business Model & Engineering Validation Roadmap

Dimension / Phase	Financial & Operational Value Proposition (For Utilities & Xylem)	Honest Engineering Boundaries & Validation Plan (TRL 3 → TRL 6)
Zero-CAPEX Deployment & Cartridge Unit Economics	Municipalities pay **$0 upfront hardware CAPEX**. Readout pods are bundled into **Evoqua** service contracts; utilities pay only a monthly Xylem Vue SaaS fee ($450/mo) plus 30-day disposable microfluidic cartridges (**$11.80 target BOM**, $45 price → 73% gross margin).	**Phase 1 (Months 1–2, TRL 3 → 4):** Cleanroom fabrication of 32-channel COC/MIP cartridges and empirical 4×4 cross-selectivity calibration against EPA Method 533 LC-MS/MS in real river/well water.
Annual Net OPEX & Hazardous Waste Savings	Extending GAC/IX media bed utilization from 68.0% to 96.4% (+28.4%) saves a medium utility (2–5 MGD) **$42,000 – $85,000/year in net OPEX** and reduces incinerated hazardous PFAS carbon waste by 28%.	**Phase 2 (Months 3–4, TRL 4 → 5):** Rapid Small-Scale Column Testing (RSSCT) under continuous flow to verify 30-day MIP/NIP biofouling stability and PINN breakthrough accuracy. **Phase 3 (Months 5–6, TRL 5 → 6):** Field pilot on a municipal GAC/IX vessel `1/4" NPT` mid-bed port.

9. PEER-REVIEWED ACADEMIC & REGULATORY REFERENCES (WITH DOI)

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